

Prove each of the statements in 23-33.

4.6.33. A necessary and sufficient condition for an integer n to be divisible by a nonzero integer d is that $n = \lfloor n/d \rfloor d$. In other words, for every integer n and nonzero integer d ,

- a. if $d \mid n$, then $n = \lfloor n/d \rfloor d$.
- b. if $\lfloor n/d \rfloor d$ then $d \mid n$.

- a. If $d \mid n$, then $n = \lfloor n/d \rfloor d$

Proof:

Suppose n and d are any integers, and $d \neq 0$.

We must show that if $d \mid n$, then $n = \lfloor n/d \rfloor d$.

Let's assume that the given statement is true.

And the definition of divisibility is:

$$d \mid n \leftrightarrow \exists k \in \mathbb{Z} \text{ such that } n = dk$$

So let some $k \in \mathbb{Z}$, and $n = dk, \forall n \in \mathbb{Z}$

$$d \mid dk = dk/d = k$$

Next we look at the definition of floor:

$$\lfloor x \rfloor = n \leftrightarrow n \in \mathbb{Z} \text{ and } n \leq x < n + 1$$

And since we know that $d \mid dk = k$, and $k \in \mathbb{Z}$, therefore $\lfloor k \rfloor = k$

And $\lfloor k \rfloor d = kd = dk$, which is what was to be shown.

□

Solution: If $d \mid n$, then $n = \lfloor n/d \rfloor d$

- b. If $\lfloor n/d \rfloor d$, then $d \mid n$

Proof:

Suppose n and d are any integers, and $d \neq 0$.

We must show that if $\lfloor n/d \rfloor d$, then $d \mid n$.

Let's assume that the given statement is true.

Let $n \in \mathbb{Z}$

Let $d \in \mathbb{Z}$

The definition of floor is:

$$\lfloor x \rfloor = n \leftrightarrow n \in \mathbb{Z} \text{ and } n \leq x < n + 1$$

Since we know both n and d are some integers, $\lfloor n/d \rfloor$ is also an integer.

Let $\lfloor n/d \rfloor$ be some $k \in \mathbb{Z}$

$$\lfloor n/d \rfloor d = kd = dk$$

The definition of divisibility is:

$$d \mid n \leftrightarrow \exists k \in \mathbb{Z} \text{ such that } n = dk$$

$n = \lfloor n/d \rfloor d = dk$, which is what was to be shown.

□

Solution: If $\lfloor n/d \rfloor d$, then $d \mid n$